ROCK, PAPER, SCISSOR GAME USING PYHTON

A report submitted in partial fulfillment of the requirements for the Award of Degree of

BACHELOR OF TECHNOLOGY
in
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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CERTIFICATE

This is to certify that the Internship report entitled **"ROCK,PAPER,SCISSOR GAME"** being submitted by **"RAYAPUDI HARSHA VARDHANI" "Y22ACM467** "is work done by him/ her and submitted during 2023– 2024 academic year, in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in CBDS&AIML,** at "**VaultofCodes in Delhi"** from **05th May, 2024** and to **10th June, 2024**

Department Internship Coordinator Head of the Department

DECLARATION

I hereby declare that the dissertation entitled **ROCK,PAPER,SCISSOR GAME** submitted for the B.Tech Degree is my work and the dissertation has not formed the basis for the award of any degree, associates, fellowship or any other similar titles.

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For successful completion of internship as an Python Programming Intern at VaultofCodes.in for a duration of 1 months starting from 5th March 2024

We found the candidate sincere, hardworking, technically sound and result oriented. We take this opportunity to appreciate his/her work and contributions to thank and wish all the best for your future endeavors

HR MANAGER
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COMPANY PROFILE and EXTERNAL GUIDE DETAILS

COMPANY PROFILE:

Vault of codes, their aim is to empower individuals with the knowlegde and skills needed to thrive with the rapidly evovling world of technology an Autonomous Scientific Society under the administrative control of Department of Electronics & Information Technology (DeitY), Ministry of Communications and Information Technology, Government of India, was set up to carry out Human Resource Development and related activities in the area of Information, Electronics & Communications Technology (IECT). It is also one of the National Examination Body, which accredits institutes/organizations for conducting courses in IT in the nonformal sector. Vault of codes, Delhi has acquired very good expertise in IT training, through its wide repertoire of causes, ranging from 'O' Level (Foundation), IT literacy courses such as CCC (Course on Computer Concept), BCC (Basic Computer Course) and other such long term and short term course in the non-formal sector like courses on Information Security, ITeSBPO(Customer Care/Banking), Computer Hardware Maintenance (CHM-O/A level), ESDM etc.

External Guide Lines:

Head of the vaultofcodes,Delhi guided me throughout the internship. She provided direction on project goals, feedback on deliverables, and support in navigating the organization's resources. Her contributions included:

• Initial project planning and scope definition

• Teaching AIML concepts and Python programming

• Regular feedback and reviews of project progress

• Offering career advice and professional development tips

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ABSTRACT:

Rock-Paper-Scissor game in python,allowing users to play against the computers. The game generates random choices for the computer and determines the winner based on the game rule's.The program uses a simple command-line interface to interact with the user,prompting them to enter their choice(rock,paper,or scissor)and displaying the result.

This project presents a simple implementation of the classic game Rock, Paper, Scissors using Python. The game is a two-player game where each player simultaneously forms one of three shapes with an outstretched hand: rock, paper, or scissors. The rules are straightforward: rock beats scissors, scissors beat paper, and paper beats rock.

Keywords: Rock Paper Scissors, hand game, user input, random number generation, conditional statements, programming basics.

PROPOSED SYSTEM:

1.Multiplayer Support: Allow two players to play against each other.

2.AI Opponent**:** Implement a basic AI opponent for single-player mode.

3.Score Tracking**:** Keep track of player scores and display them in the Game Interface**.**

4.Game Statistics: Display game statistics, such as win/loss ratio and most played moves.

5.Customization Options: Allow players to customize game settings, such as game mode and AI difficulty**.**

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1. Introduction:

Rock Paper Scissors is a timeless and universal hand game that has been enjoyed by people of all ages and cultures for centuries. This simple yet strategic game is a great way to pass the time, challenge friends and family, and even develop critical thinking skills.

Objective:

The objective of Rock Paper Scissors is to outsmart your opponent by choosing one of three hand signals: Rock, Paper, or Scissors. Each signal has a unique relationship with the others, creating a cyclical pattern of winning and losing combinations**.**

How to Play:

1. Players simultaneously throw one of three hand signals: Rock (a closed fist), Paper (a flat hand), or Scissors (a "V" shape).

2. The winning hand beats the losing hand as follows:

- Rock beats Scissors (crushes them)
- Scissors beats Paper (cuts it)
- Paper beats Rock (covers it)

3. If both players throw the same signal, the game is a tie.

Get Ready to Play!

In this digital version of Rock Paper Scissors, you'll play against the computer. Will you emerge victorious or fall to defeat? There's only one way to find out! Let's start playing!

2.Existing System For Rock,Paper,Scissor Game :

Description :

A simple game where two players choose on of rock , paper, or scissors and the winner id determined by the game's rule.

Existing features:

1)User input: players enter their choice(rock, paper, or scissors).

2)Computer choice: The System generates a random choice foe the computer playgame

3)game logic: The system determines the winner based on the game's rules.

4)Result display: The system displays the result , including the winner and the choices made.

System Flow:

1. User inputs their choice (Rock, Paper, or Scissors).
2. Computer generates a random choice.
3. Game logic determines the winner based on the rules.
4. Result is displayed to the user.
5. User can play again or quit.

Existing Implementations:

1. Online Rock Paper Scissors games (e.g., web-based, mobile apps).

2. Command-line Rock Paper Scissors games (e.g., Python, Java, C++).

3. GUI-based Rock Paper Scissors games (e.g., Python with Tkinter, Java with Swing).

Features:

1. Single-player mode against the computer.

2. Multi-player mode for playing against other users.

3. Score tracking and display.

4. Option to play best-of-three or best-of-five series.

5. Help or instructions for new players.

This existing system provides a basic structure for a Rock Paper Scissors game, which can be enhanced or modified to add new features or functionality.

These existing systems demonstrate various implementations of the Rock Paper Scissors game, including web-based, mobile apps, command-line, GUI-based, and online platforms. They can serve as inspiration or starting points for building your own Rock Paper Scissors game.

Python Rock Paper Scissors: A command-line game built with Python, where users can play against the computer.

CodePen Rock Paper Scissors: A web-based game built with HTML, CSS, and JavaScript, hosted on CodePen.

GamezArena Rock Paper Scissors: A web-based game where users can play against other players in real-time

3.Software Requirements For Rock,Paper,Scissor Game

Functional Requirements:

1.User Input :Allow users to input their choice (Rock, Paper, or Scissors).

2.Computer Choice:Generate a random choice for the computer.

3.Game Logic: Determine the winner based on the game rules.

4. Result Display: Display the user's choice, computer's choice, and the result.

5. Play Again: Allow users to play again or quit.

6. Score Tracking: Keep track of the user's score (optional).

Non-Functional Requirements:

1. Performance: Respond to user input within 1 second.

2. Usability: Provide a user-friendly interface for input and output.

3. Reliability: Ensure the game logic is correct and consistent.

4. Security: Protect against cheating or tampering.

5. Compatibility: Run on multiple platforms (e.g., Windows, macOS, Linux).

System Requirements:

1. Operating System: Windows, macOS, or Linux.

2. Programming Language**:** Python, Java, JavaScript, or C++**.**

3. Memory**:** 128 MB RAM or more.

4. Storage**:** 10 MB disk space or more**.**

5. Display**:** 800x600 resolution or higher**.**

Software Dependencies:

1. Random number generator library (e.g., random in Python).

2. User interface library (e.g., Tkinter in Python).

3. Optional: score tracking library or database**.**

Assumptions and Constraints:

1. The game is played between a single user and the computer.

2. The game follows the traditional Rock Paper Scissors rules.

3. The user input is valid (Rock, Paper, or Scissors).

4. The computer choice is random and unbiased**.**

4.Software Design For Rock,Paper,Scissor Game

Architecture:

1. User Interface (UI) Layer:

- Handles user input and output
- Displays game results and scores

2. Game Logic Layer:

- Determines the winner based on game rules
- Generates computer's random choice

3. Data Storage Layer**:**

- Stores user's score and game history (optional)

Components:

1. GameController:

- Coordinates UI and game logic interactions
- Manages game state and flow

2. UserInputHandler:

- Handles user input (Rock, Paper, or Scissors)
- Validates user input

3. ComputerPlayer:

- Generates random choice (Rock, Paper, or Scissors)

4. GameRules:

- Defines winning conditions and game rules

5. ScoreTracker:

- Tracks user's score and game history (optional)

Interfaces:

1. UserInterface:

- Displays game results and scores
- Provides input mechanism for user

2. GameLogicInterface:

- Provides access to game rules and logic

Classes:

1. GameController:

- Attributes: game state, user score, game history
- Methods: start game, handle user input, determine winner, update score

2. UserInputHandler:

- Attributes: user input
- Methods: handle input, validate input

3. ComputerPlayer:

- Attributes: random choice
- Methods: generate random choice

4. GameRules:

- Attributes: winning conditions
- Methods: determine winner

5. ScoreTracker:

- Attributes: user score, game history

- Methods: update score, retrieve score

Sequence Diagram:

1. User inputs choice (Rock, Paper, or Scissors)

2. UserController handles input and validates

3. GameController determines winner using GameRules

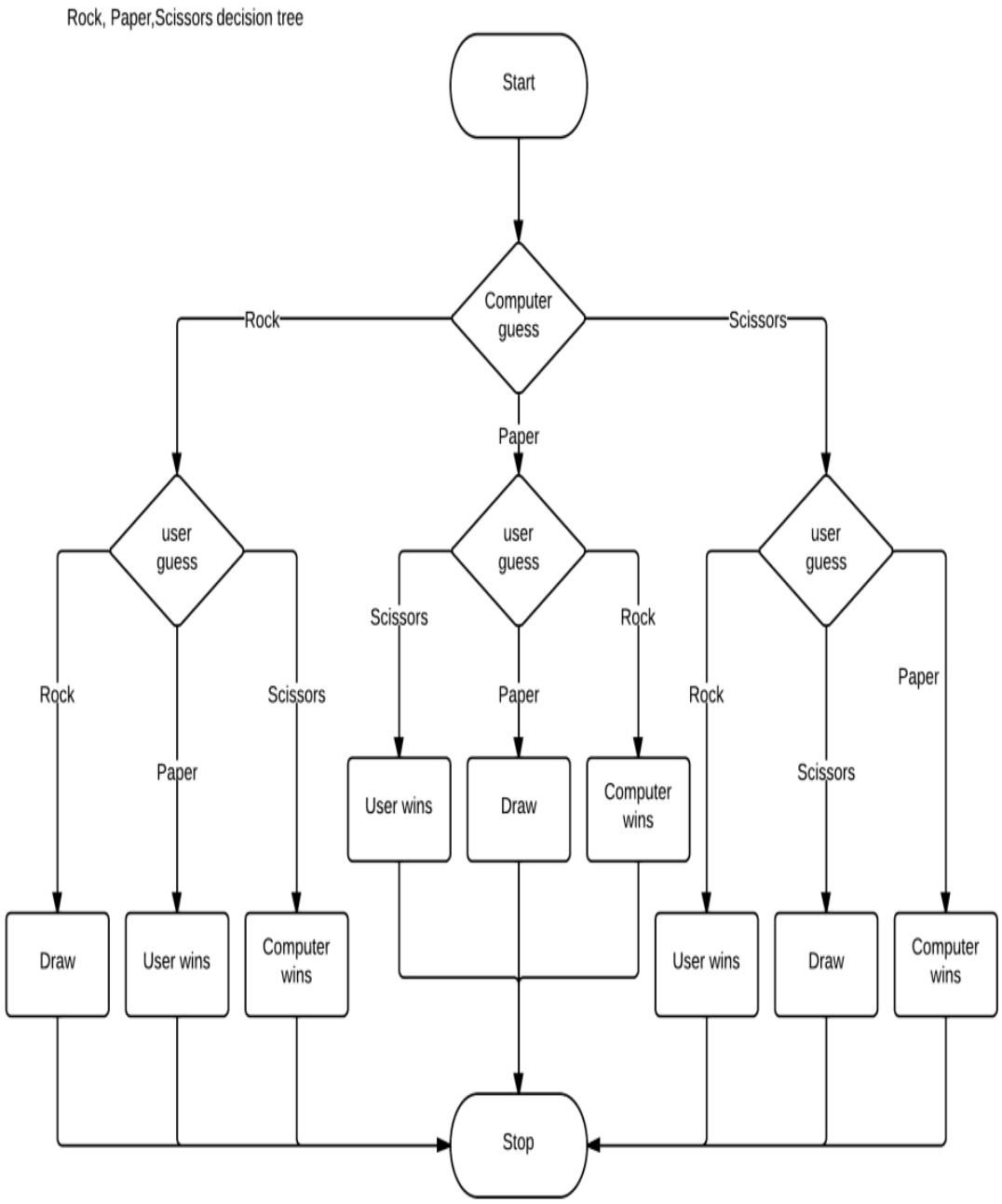
4. ComputerPlayer generates random choice

5. GameController updates score using ScoreTracker

6. UserController displays game results and scores

This design provides a clear separation of concerns, making it easy to maintain and extend the game. The GameController coordinates the game flow, while the UserInputHandler and ComputerPlayer handle user input and computer choice, respectively. The GameRules and ScoreTracker provide access to game logic and score tracking.

Flow Chart:



5.Technology and Methodology For Rock,Paper,Scissor Game

Designing and developing a rock paper scissor game involves choosing appropriate technologies and methodologies to ensure a functional, engaging, and user-friendly experience. Here's a detailed look at the technology stack and methodologies you might use:

Technology Stack

1. Front-End Technologies

- **Web-Based Development:**

  HTML/CSS: For structuring and styling the user interface.

  JavaScript: For implementing game logic and interactions.

  Frameworks/Libraries:

  - **React.js:** For building dynamic and interactive user interfaces.

2. Back-End Technologies (for online versions)

- **Server-Side Languages:**
  - **Node.js:**

    For handling server-side logic and real-time interactions.

  - **Python (Flask/Django):**

    For building robust server-side applications and APIs.

- **Ruby on Rails:**

  For developing web applications quickly and efficiently.

3. **Development Environment**:

- **Text Editor or IDE**:

  Code editors like Visual Studio Code, PyCharm, or even simple text editors such as Notepad++ are used to write and modify the Hangman game code.

- **Terminal/Command Line**:

  Used to execute the Python script and interact with the game.

4. **Version Control**:

- **Git**:

  For tracking changes and managing versions of the code, Git can be used. Platforms like GitHub or GitLab host repositories and facilitate collaboration

5.Libraries:

Standard Library:

The example uses Python's built-in libraries (random for word selection and basic string operations). No additional external libraries are required for this simple implementation.

Methodologies

1. Agile Development

- **Iterative Approach:**

  Develop the game in iterative cycles, allowing for regular feedback and adjustments.

- **Scrum/Kanban:**

  Utilize Scrum for managing tasks in sprints or Kanban for a more continuous workflow.

2.User-Centered Design

- **User Research:**

  Understand the target audience's needs and preferences through surveys, interviews, or usability testing.

- **Prototyping:**

  Create wireframes and prototypes to visualize and test the game's interface and interactions before full development.

3.Testing:

- **Unit Testing**: Test individual functions to ensure they work correctly. For example, test word selection, input validation, and state updating.
- **Playtesting**: Run the game multiple times to check for bugs and ensure that it handles all expected scenarios, including edge cases like invalid inputs and repeated guesses.

4.Documentation:

- **Code Comments**:

  Provide inline comments in the code to explain logic and functionality.

- **User Guide**:

  Create a simple guide or README file explaining how to run the game and its rules.

6.Coding For Rock Paper Scissor Game

```
import random
def play_game():
    while True:
        user_choice = input("Enter your choice (rock/paper/scissors) or 'q' to
quit:").lower()
        if user_choice == 'q':
            break
        if user_choice not in ['rock', 'paper', 'scissors']:
            print("Invalid choice. Please enter rock, paper, or scissors.")
            continue
         possible_choices = ['rock', 'paper', 'scissors']
        computer_choice = random.choice(possible_choices)
        print(f"\nYou chose {user_choice}, computer chose {computer_choice}.\n")
        if user_choice == computer_choice:
            print(f"Both players selected {user_choice}. It's a tie!")
        elif user_choice == 'rock':
            if computer_choice == 'scissors':
                print("Rock smashes scissors! You win!")
            else:
                print("Paper covers rock! You lose.")
        elif user_choice == 'paper':
            if computer_choice == 'rock':
                print("Paper covers rock! You win!")
            else:
                print("Scissors cuts paper! You lose.")
        elif user_choice == 'scissors':
            if computer_choice == 'paper':
                print("Scissors cuts paper! You win!")
            else:
                print("Rock smashes scissors! You lose.")

play_game()
```

Output :

```
Enter your choice (rock/paper/scissors) or 'q' to quit:  paper

You chose paper, computer chose paper.

Both players selected paper. It's a tie!
Enter your choice (rock/paper/scissors) or 'q' to quit:  scissors

You chose scissors, computer chose scissors.

Both players selected scissors. It's a tie!
Enter your choice (rock/paper/scissors) or 'q' to quit:  rock

You chose rock, computer chose scissors.

Rock smashes scissors! You win!
Enter your choice (rock/paper/scissors) or 'q' to quit:  q
```

Code Explanation :

1. Importing the random module: We import the random module to generate the computer's random choice.

2. Defining the play_game function: We define a function play_game that contains the game logic.

3. Game loop: The game runs in an infinite loop until the user chooses to quit.

4. Getting user input: We ask the user to enter their choice (rock, paper, or scissors) or 'q' to quit. We convert the input to lowercase using the lower() method.

5. Validating user input: We check if the user input is valid (rock, paper, or scissors). If not, we display an error message and continue to the next iteration.

6. Generating computer choice: We generate the computer's random choice using the random.choice() function.

7. Determining the winner: We compare the user choice and computer choice to determine the winner based on the game rules.

8. Displaying the result: We display the result of the game, including the computer's choice and the winner.

9. Asking user to play again: We ask the user if they want to play again. If they enter 'q', we break out of the loop and end the game.

Running the game

To run this Hangman game:

1. Copy the code into a file named rpg.py.
2. Open a terminal or command prompt.
3. Navigate to the directory containing rpg.py.
4. Run the script using the command python rpg.py.

7.Result and discussion:

Result:

Victory:

- If the user chooses Rock and the computer chooses Scissors, the user wins.
- If the user chooses Scissors and the computer chooses Paper, the user wins.
- If the user chooses Paper and the computer chooses Rock, the user wins.
- If both the user and the computer choose the same option, the result is a tie.

Output:

Enter your choice (rock/paper/scissors) or 'q' to quit: rock

You chose rock, computer chose scissors.

Rock smashes scissors! You win!

Enter your choice (rock/paper/scissors) or 'q' to quit: paper

You chose paper, computer chose scissors.

Scissors cuts paper! You lose.

Enter your choice (rock/paper/scissors) or 'q' to quit: scissors

You chose scissors, computer chose scissors.

Both players selected scissors. It's a tie!

Enter your choice (rock/paper/scissors) or 'q' to quit:q

Discussion:

1. Overview

Number of Players: 2 (User and Computer)

Objective: To win by choosing a hand signal that beats the opponent's signal.

Hand Signals:

1. Rock: A closed fist

2. Paper: A flat hand with fingers together

3. Scissors: A "V" shape made with the index and middle fingers

Gameplay:

1. The user and computer simultaneously choose a hand signal.

2. The winner is determined by the following rules:

   - Rock beats Scissors (crushes them)

   - Scissors beats Paper (cuts it)

   - Paper beats Rock (covers it)

3. If both players choose the same signal, the result is a tie.

Game Modes:

1. Single Round: Play a single round against the computer.

2. Best of Three: Play a best-of-three series against the computer.

Benefits:

1. Improves Strategic Thinking: Encourages players to think strategically and make informed decisions.

2. Enhances Hand-Eye Coordination: Develops hand-eye coordination and fine motor skills.

3. Promotes Social Interaction: Encourages social interaction and friendly competition.

2. Educational Value

1. Decision Making: Encourages critical thinking and decision-making skills.

2. Strategy Development: Fosters strategic planning and problem-solving abilities.

3. Probability and Statistics: Introduces basic probability concepts and statistical analysis.

4. Pattern Recognition: Enhances pattern recognition and analytical skills.

5. Social Skills: Develops social interaction, communication, and good sportsmanship.

6. Emotional Intelligence: Teaches emotional management, self-awareness, and self-regulation.

7. Risk Management: Encourages risk assessment, weighing options, and making informed decisions.

8. Adaptability and Flexibility: Promotes adaptability, flexibility, and adjusting to new situations.

9. Critical Thinking: Encourages critical thinking, analysis, and evaluation of information.

10. Mathematical Concepts: Introduces basic mathematical concepts, such as probability, statistics, and patterning.

11. Science and Engineering: Relates to scientific and engineering principles, like game theory and algorithmic thinking.

12. Language and Communication: Enhances language skills, communication, and expression of thoughts and ideas.

13. Cultural Awareness: Exposes players to different cultural variations and rules of the game.

14. Historical Context: Provides insight into the game's history, evolution, and significance.

15. Life Skills: Develops essential life skills, such as winning graciously, losing with dignity, and perseverance.

8.Conclusion

In conclusion, the Rock Paper Scissors game is a timeless and universal game that offers a unique combination of simplicity, strategy, and social interaction. Through its simple yet engaging gameplay, Rock Paper Scissors:

- Develops critical thinking, problem-solving, and decision-making skills
- Introduces basic probability and statistical concepts
- Fosters social skills, emotional intelligence, and good sportsmanship
- Encourages strategic planning, adaptability, and risk management
- Provides a fun and engaging way to learn and develop essential skills

Whether played in a casual or competitive setting, Rock Paper Scissors remains a beloved game that transcends age, culture, and geography. Its enduring

popularity is a testament to its ability to bring people together, promote social interaction, and provide a shared experience that is both enjoyable and enriching.

In the end, Rock Paper Scissors is more than just a game - it's a cultural phenomenon that continues to inspire, educate, and entertain people around the world.

9.Reference:

1. "The Official Rock Paper Scissors Strategy Guide" by Douglas Walker and Graham Walker

2. "Rock Paper Scissors: The Ultimate Guide" by Jason Simmons

3. "The Rock Paper Scissors Handbook" by John McCarthy

Articles:

1. "The Science of Rock Paper Scissors" by Scientific American

2. "The Mathematics of Rock Paper Scissors" by The Guardian

Key References:

1.Educational Resources:

- *"Python Programming: An Introduction to Computer Science"* by John Zelle.

  This textbook is widely used in computer science education and includes a variety of programming exercises, including Hangman.

- *"Automate the Boring Stuff with Python"* by Al Sweigart. This book provides practical programming exercises and includes a detailed example of a Hangman game.

2. **Online Tutorials:**

- Codecademy Python Course: Offers interactive Python courses that often include projects like rock paper scissor to help learners apply their skills.
- Real Python: Provides tutorials and articles on Python programming,
- including examples and exercises related to rock paper scissor.

3.Programming Communities:

Stack Overflow: A popular platform where programmers discuss and share code, including implementations of rock paper scissor.

GitHub: Hosts numerous repositories with rock paper scissor game implementations in various programming languages, serving as a reference for coding practices.

These resources offer a broad overview of rock paper scissor role in learning programming and provide practical examples for those interested in understanding or expanding on the basic game